



Jae U YOON

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"Accelerating the future through innovative research"

UPDATED ON July 1st, 2026

EDUCATION

UNIVERSITY OF ULSAN

Mar. 2021 ~ Feb. 2027 (Expected)

B.S. IN SCHOOL OF MATERIALS SCIENCE & ENGINEERING

- Total GPA of 3.83 / 4.5

(Credits taken: 129/137 as of 7th semesters)

YONSEI UNIVERSITY (Academic Exchange)

Mar. 2022 ~ Aug. 2022

B.S. IN SCHOOL OF MATERIALS SCIENCE & ENGINEERING

INTERNSHIP

ADVANCED ELECTRO-CHEMISTRY LAB (Prof. Sun-jeong Kim)

Sep. 2022 ~ Feb. 2026

UNDERGRADUATED RESEARCHER

<Personal research project>

Mar. 2025 ~ Feb. 2026

- One-step electrodeposition of $\text{Cu}_2\text{MnSnS}_4$ thin films on FTO glass for solar cell applications

<Government-funded research project>

Sep. 2022 ~ Feb. 2023

- Study on the electro-refining process for separation and recovery of valuable metals in waste resources

ADVANCED RECHARGEABLE BATTERIES LAB (Prof. Hyun-jeong Lee)

Jul. 2026 ~

INTERN

AWARDS & HONORS

-AWARDS-

2026. 06 1st place, Capstone award competition (Prof. Hyung-cheol Shim)

Simulation researcher

"Super-P/Graphene composite conductive additive research for high-capacity battery development"

2026. 06 2nd place, Capstone design competition (Prof. Hyung-cheol Shim)

Simulation researcher

"Super-P/Graphene composite conductive additive research for high-capacity battery development"

2025. 12 2nd place, Capstone design competition (Prof. Eun-chae Jeon)

Team leader & Presenter

"Reliability evaluation and lifetime modeling of foldable displays"

-HONORS-

2025. 09 Semester High Honor, Awarded to students with high achievements throughout the semester

2023. 03 Semester High Honor, Awarded to students with high achievements throughout the semester

2022. 03 Academic Exchange Scholarship, Awarded to selected students for active participation in the academic exchange program (Yonsei University)

2021. 09 Semester High Honor, Awarded to students with high achievements throughout the semester

SKILLS

English Proficiency

- 2025. 12 – OPIC IH

Certification

- 2025.01 – Six-Sigma Green Belt

Experimental Equipment

- Potentiostat / Galvanostat, SEM-EDS, XRD, Electric Furnace, AFM, UV-Vis, AAS, Four-Point Probe

Office Automation

- Microsoft software (Word, PowerPoint, Excel), Origin Pro

Development Tool

- Visual Studio code, Visual Studio, Antigravity, MATLAB

3D Modeling Tool

- Autodesk AutoCAD

ADDITIONAL INFORMATION

Personal Project

- Developed and visualized a computational simulation using Antigravity with Claude Code and Gemini Code Assist to analyze the effect of Super-P/Graphene composite conductive additive

- Developed and visualized a computational simulation using Visual Studio code with Claude Code and Gemini Code Assist to analyze the effect of electrolyte agitation on the diffusion layer thickness within the Electrical Double Layer (EDL)
- Designed and machined a custom Teflon housing to standardize the exposed area of FTO glass during the electrodeposition process
- Designed and 3D-printed a multi-slot custom holder to enable the batch ultrasonic cleaning of multiple FTO glass substrates

ACTION PLAN

- Summer 2026: Internship at Advanced Rechargeable Batteries Lab(Prof. Hyeon-jeong Lee, UNIST)